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2-31 In Figure P2-31 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

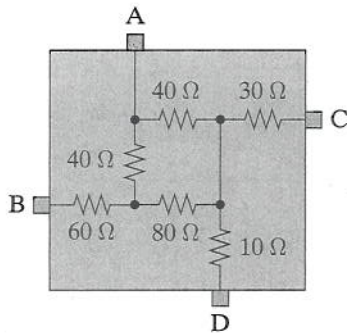


FIGURE P2-31

2-32 Select a value of R_L in Figure P2-32 so that $R_{EQ} = 25\text{ k}\Omega$.

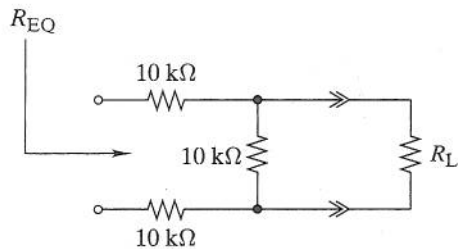


FIGURE P2-32

2-33 Repeat Problem 2-32 for $R_{EQ} = 15\text{ k}\Omega$. Caution: R_L must be positive.

2-34 Find the equivalent practical voltage source at terminals A and B in Figure P2-34.

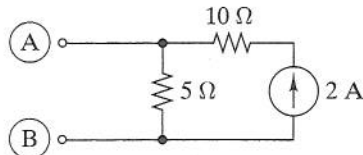


FIGURE P2-34

2-35 In Figure P2-35 the $i-v$ characteristic network N is $v + 50i = 5\text{ V}$. Find the equivalent practical current source for the network.

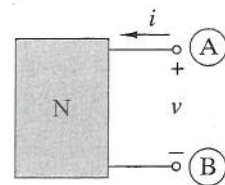


FIGURE P2-35

2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60\text{ k}\Omega$.

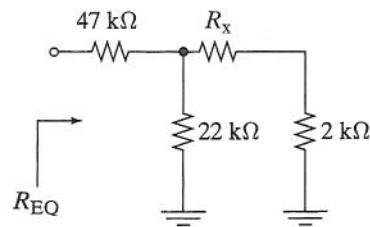


FIGURE P2-36

2-37 Show how to interconnect standard $3.9\text{-k}\Omega$ resistors to obtain equivalent resistances of $1\text{ k}\Omega \pm 5\%$, $5\text{ k}\Omega \pm 5\%$, and $10\text{ k}\Omega \pm 5\%$.

2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

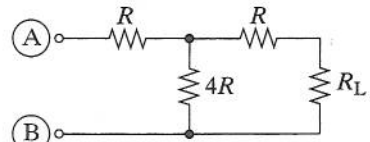


FIGURE P2-38

2-39 What is the range of R_{EQ} in Figure P2-39?

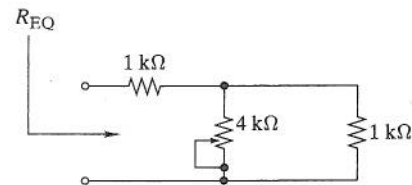


FIGURE P2-39

2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.

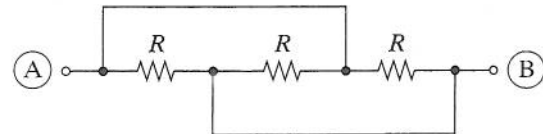


FIGURE P2-40

OBJECTIVE 2-5 VOLTAGE AND CURRENT DIVISION (SECT. 2-5)

- (a) Given a circuit with elements connected in series or in parallel, use voltage or current division to find specified voltages or currents.
 - (b) Design a voltage or current divider that delivers specified output signals within stated constraints.
- See Examples 2-13, 2-14, 2-16, 2-17, 2-18 and Exercises 2-10, 2-11, 2-12

- 2-41 Use voltage division in Figure P2-41 to obtain an expression for v_L in terms of R , R_L , and v_S .

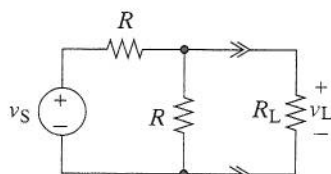


FIGURE P2-41

- 2-42 Use current division in Figure P2-42 to obtain an expression for v_L in terms of R , R_L , and i_S .

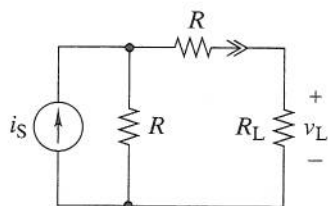


FIGURE P2-42

- 2-43 Find i_x in Figure P2-43.

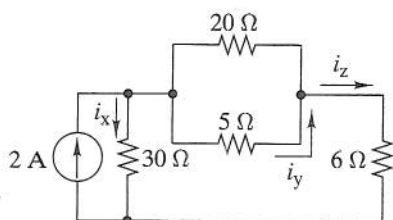


FIGURE P2-43

- 2-44 Find i_y and i_z in Figure P2-43.

- 2-45 Find the range of values of v_O in Figure P2-45.

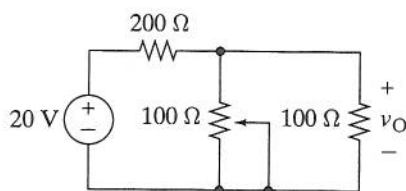


FIGURE P2-45

- 2-46 Figure P2-46 shows a resistance divider connected in a general circuit.

- (a) What is the relationship between v_1 and v_2 when $i_1 = 0$?

- (b) What is the relationship between v_1 and v_2 when $i_2 = 0$?

- (c) What is the relationship between i_1 and i_2 when $v_1 = 0$?

- (d) What is the relationship between i_1 and i_2 when $v_2 = 0$?

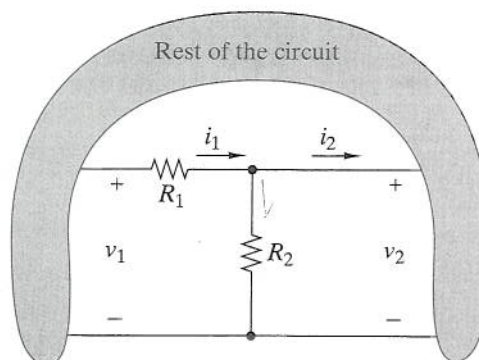


FIGURE P2-46

- 2-47 **D** Figure P2-47 shows an ammeter circuit consisting of a D'Arsonval meter, a two-position selector switch, and two shunt resistors. A current of 0.5 mA produces full-scale deflection of the D'Arsonval meter, whose internal resistance is $R_M = 50 \Omega$. Select the shunt resistance R_1 and R_2 so that $i_x = 10$ mA produces full-scale deflection when the switch is in position A, and $i_x = 50$ mA produces full-scale deflection when the switch is in position B.

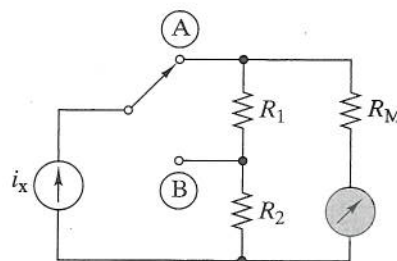


FIGURE P2-47

- 2-48 **D** Select values for R_1 , R_2 , and R_3 in Figure P2-48 so the voltage divider produces the two output voltages shown.

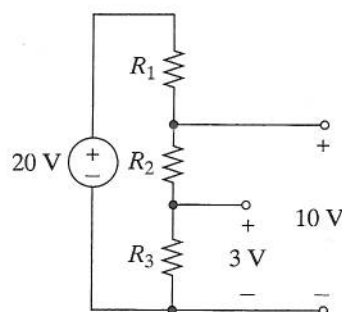


FIGURE P2-48

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when $v_2 = 0$?

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- 2-49 **D** Select a value of R_x in Figure P2-49 so that $v_L = 3$ V.

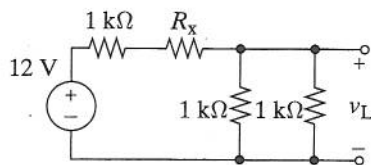


FIGURE P2-49

- 2-50 **D** Select a value of R_x in Figure P2-50 so that $v_L = 6$ V. *Caution:* R_x must be positive.

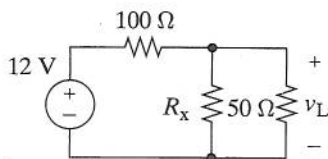


FIGURE P2-50

OBJECTIVE 2-6 Circuit Reduction (SECT. 2-6)

Given a circuit consisting of linear resistors and an independent source, find selected signal variables using successive application of series/parallel equivalence, source transformations, and voltage/current division. See Example 2-20, 2-21, 2-22, 2-23 and Exercises 2-13, 2-14, 2-15

- 2-51 Use circuit reduction to find v_x and i_x in Figure P2-51.

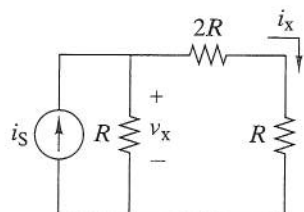


FIGURE P2-51

- 2-52 Use circuit reduction to find i_x in Figure P2-52.

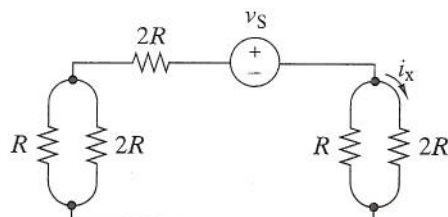


FIGURE P2-52

- 2-53 Use circuit reduction to find v_x in Figure P2-53.

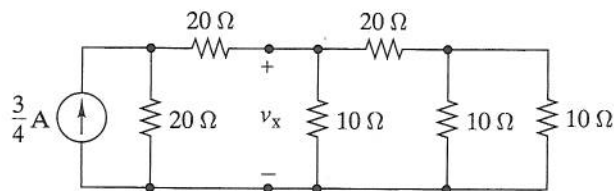


FIGURE P2-53

- 2-54 Use circuit reduction to find v_x and i_x in Figure P2-54.

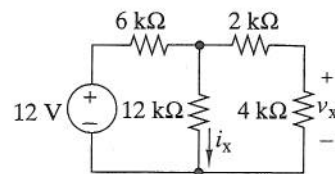


FIGURE P2-54

- 2-55 Use source transformation to find i_x in Figure P2-55.

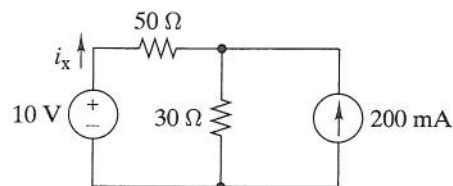


FIGURE P2-55

- 2-56 Use source transformation to find i_x in Figure P2-56.

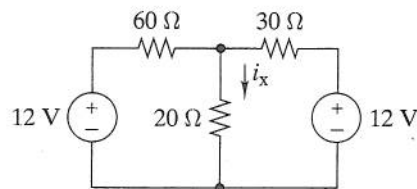


FIGURE P2-56

- 2-57 Use source transformations in Figure P2-57 to relate v_O to v_1 , v_2 , and v_3 .

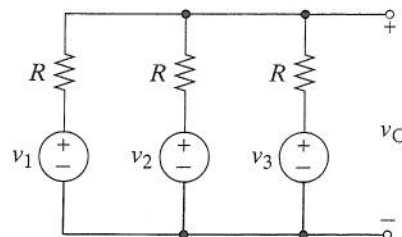


FIGURE P2-57

- 3-30 Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-30.

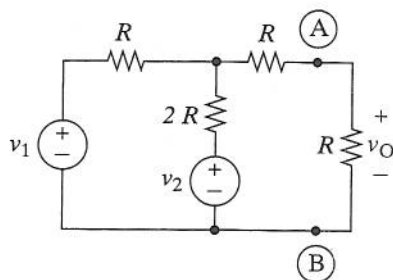


FIGURE P3-30

- 3-31 Use the superposition principle to find v_O in terms of i_1 , i_2 , and R in Figure P3-31.

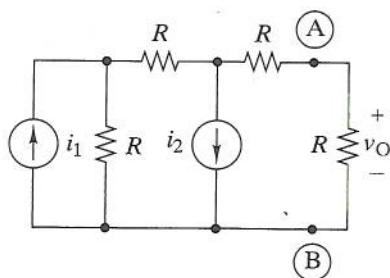


FIGURE P3-31

- 3-32 A linear circuit containing two sources drives a $100\text{-}\Omega$ load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. *Hint:* The answer is not 4.25 W . Why?

- 3-33 A linear circuit is driven by an independent voltage source $v_s = 10\text{ V}$ and an independent current source $i_s = 10\text{ mA}$. The output voltage is $v_O = 2\text{ V}$ when the voltage source is on and the current source off. The output is $v_O = 1\text{ V}$ when both sources are on. Find the output voltage when $v_s = 5\text{ V}$ and $i_s = -10\text{ mA}$.

- 3-34 A certain linear circuit has three input voltages and one output voltage v_O . The following table lists the output for different values of the three inputs. Find the input-output relationship for the circuit.

$v_{S1}(\text{V})$	$v_{S2}(\text{V})$	$v_{S3}(\text{V})$	$v_O(\text{V})$
2	4	-4	1
1	2	2	1.5
1	4	2	2

- 3-35 This problem involves designing a resistive circuit with two inputs v_{S1} and v_{S2} and a single output voltage v_O . Design the circuit so that $v_O = K(v_{S1} + 4v_{S2})$ is delivered across a $500\text{-}\Omega$ load. The value of K is not specified but should be greater than $1/10$.

OBJECTIVE 3-3 THÉVENIN AND NORTON EQUIVALENT CIRCUITS (SECT. 3-4)

Given a circuit containing linear resistors and independent sources:

- Find the Thévenin or Norton equivalent at a specified pair of terminals.
- Use the Thévenin or Norton equivalent to find the signals delivered to linear or nonlinear loads.

See Examples 3-13, 3-14, 3-15, 3-16 and Exercises 3-16, 3-17, 3-18

- 3-36 (a) Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.

- (b) Use the equivalent circuit found in part (a) to find i_L in terms of i_s , R_1 , R_2 , and R_L .

- (c) Check your answer in part (b) using current division.

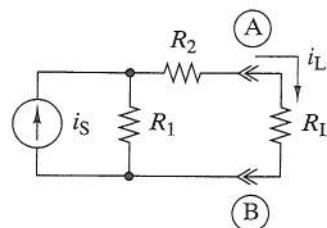


FIGURE P3-36

- 3-37 Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5\text{ }\Omega$, $10\text{ }\Omega$, and $50\text{ }\Omega$.

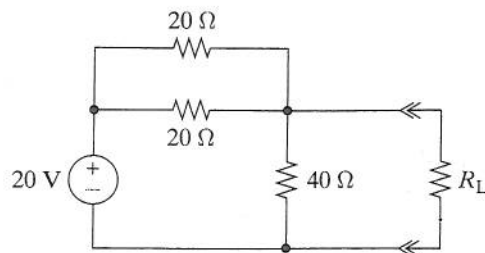


FIGURE P3-37

- 3-38 Find the Norton equivalent seen by R_L in Figure P3-38. Find the current through the load when $R_L = 6\text{ }\Omega$, $12\text{ }\Omega$, and $60\text{ }\Omega$.

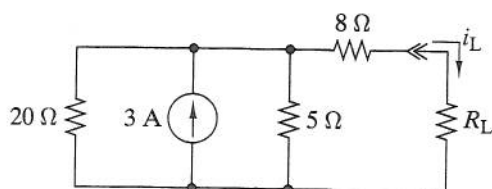


FIGURE P3-38

- 3-39 Find the Thévenin equivalent seen by R_L in Figure P3-39. Find the power delivered to the load when $R_L = 50 \text{ k}\Omega$ and $200 \text{ k}\Omega$.

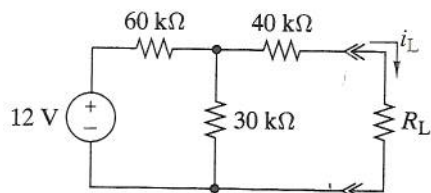


FIGURE P3-39

- 3-40 Find the Norton equivalent at terminals A and B in Figure P3-40.

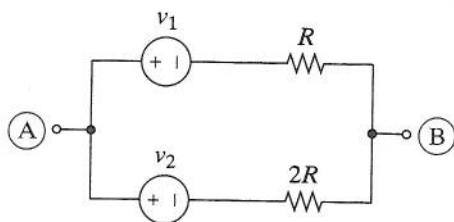


FIGURE P3-40

- 3-41 The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_x in Figure P3-41. Find the Thévenin equivalent circuit seen looking to the left of terminals A and B. Find the Thévenin equivalent circuit seen looking to the right of terminals A and B. Connect these equivalent circuits together and find the voltage v_x .

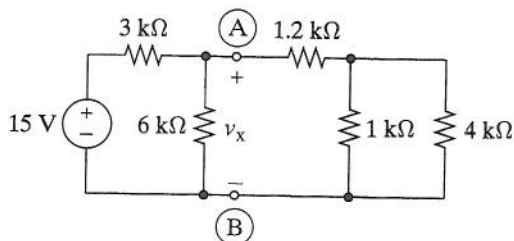


FIGURE P3-41

- 3-42 Figure P3-42 shows an active circuit with two accessible terminals. The output current is $i = 10 \text{ mA}$ when $v = 0$. The output voltage is $v = 6 \text{ V}$ when a $2.4 \text{ k}\Omega$ resistor is connected between the terminals. How much current would this source deliver to a 6-V battery?

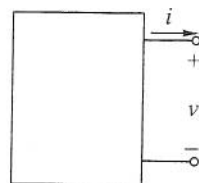


FIGURE P3-42

- 3-43 The $i-v$ characteristic of the active circuit in P3-42 is $5v + 500i = 60$. Find the output voltage when a $500\text{-}\Omega$ resistive load is connected across the two accessible terminals.

- 3-44 Figure P3-42 shows a source circuit with two accessible terminals. Some voltage and current measurements at the accessible terminals are

$v(\text{V})$	-10	-5	0	+5	+10	12	13	14
$i(\text{mA})$	+5	+4	+3	+2	+1	0	-1	-2

- Use these data to plot the source $i-v$ characteristic.
- Develop a Thévenin equivalent circuit valid on the range $|v| < 10 \text{ V}$.
- Use the equivalent circuit to predict the source v_{oc} and i_{sc} .
- Compare your results in part (c) with the given measurements and explain any differences.

- 3-45 The Thévenin equivalent parameters of a practical voltage source are $v_T = 25 \text{ V}$ and $R_T = 150 \text{ }\Omega$. Find the smallest load resistance for which the load voltage exceeds 15 V .

- 3-46 Use a sequence of source transformations to find the Thévenin equivalent at terminals A and B in Figure P3-46.

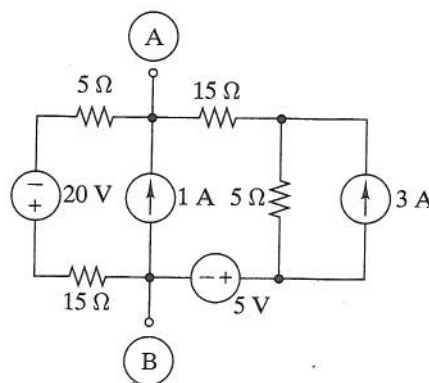


FIGURE P3-46

2-58 The current through R_L in Figure P2-58 is 40 mA. Use source transformations to find R_L .

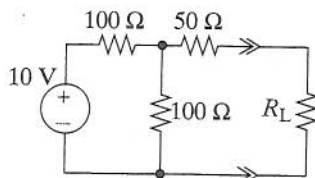


FIGURE P2-58

2-59 The voltage across R_L in Figure P2-58 is 2.5 V. Use source transformations to find R_L .

2-60 The box in the circuit in Figure P2-60 is a resistor whose value can be anywhere between $8\text{ k}\Omega$ and $80\text{ k}\Omega$. Use circuit reduction to find the range of values of v_x .

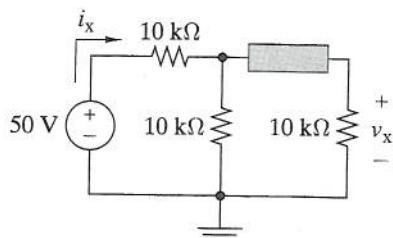


FIGURE P2-60

INTEGRATING PROBLEMS

2-61 Nonlinear Device Characteristics

The circuit in Figure P2-61 is a parallel combination of a $50\text{-}\Omega$ linear resistor and a varistor whose $i-v$ characteristic is $i_v = 2.6 \times 10^{-5}v^3$. For small voltage the varistor current is quite small compared to the resistor current. For large voltages the varistor dominates because its current increases more rapidly with voltage.

- Plot the $i-v$ characteristic of the parallel combination.
- State whether the parallel combination is linear or nonlinear, active or passive, and bilateral or nonbilateral.
- Find the range of voltages over which the resistor current is at least 10 times as large as the varistor current.
- Find the range of voltages over which the varistor current is at least 10 times as large as the resistor current.

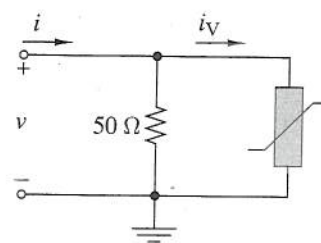


FIGURE P2-61

2-62 Center Tapped Voltage Divider

Figure P2-62 shows a voltage divider with the center tap connected to ground. Derive equations relating v_A and v_B to v_S , R_1 , and R_2 .

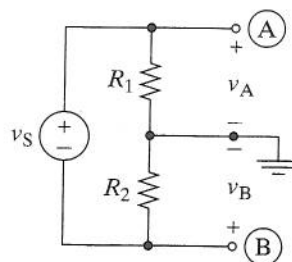


FIGURE P2-62

2-63 Combined $i-v$ Characteristics

Figure P2-63 shows an interconnection of practical voltage source and a two-terminal network N. The $i-v$ relationship of the network N is $i_1 = (0.01v_1 + 3)\text{ A}$. Find the equivalent practical voltage source of the combination at terminals A and B.

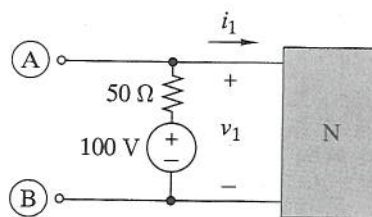


FIGURE P2-63

2-64 Programmable Voltage Divider

Figure P2-64 shows a programmable voltage divider in which digital inputs b_0 and b_1 control complementary analog switches connecting a multitap voltage divider to the analog output v_o . The switch positions in the figure apply when digital inputs are low. When inputs go high, the switch positions reverse. Find the analog output voltage for $(b_1, b_0) = (0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$ when $V_{\text{REF}} = 12\text{ V}$.